

System overview



Universal brand mark
#25856A
Logo / mark across light and dark contexts



UI brand primary
#165E4A
Primary action in the light theme

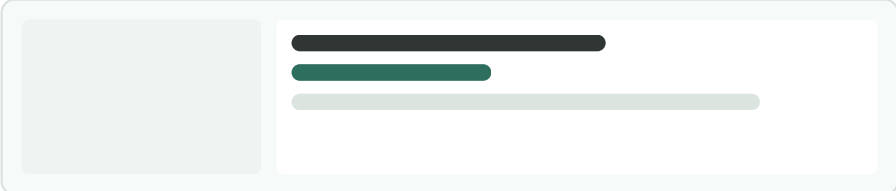


Secondary accent
#2E8B6E
Secondary brand and decorative accent



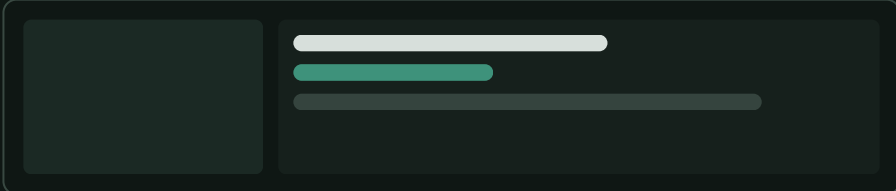
Warm accent
#D9A441
Sparse emphasis and data visualization

Light theme



Canvas #F6FAF8 · Text #18201D · Primary #165E4A

Dark theme



Canvas #0F1714 · Text #EDF5F1 · Primary #43A086

Monochrome theme



Canvas #FFFFFF · Text #111111 · Accent #111111

Semantic status roles

 **Success**

 **Info**

 **Warning**

 **Error**

 **Progress**

 **Paused**

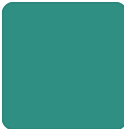
 **Cancelled**

 **Neutral**

Data categorical palette



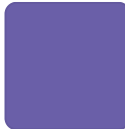
data-01 Pine
#1B6F5A



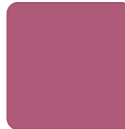
data-02 Teal
#2F8F83



data-03 Blue
#3A78B8



data-04 Violet
#6A5FA8



data-05 Rose
#B05A7A



data-06 Terracotta
#C56A3A



data-07 Ochre
#C19428



data-08 Olive
#66854A

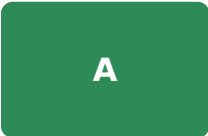


data-09 Slate teal
#4F7C82



data-10 Taupe
#7A6D61

Canonical nucleotide colors



A
#2E8B57



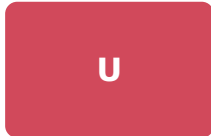
C
#2F6FDB



G
#D99000



T
#D1495B



U
#D1495B

T and U share a color but always retain their letter label.

Brand assets - reserved locations

Full mark
pending approved artwork

Micro mark
pending approved artwork

[mark] JELICA
horizontal lockup - pending

.jelica document icon
pending approved artwork

1. Brand foundation and token architecture

A pine-derived identity without turning the whole interface green.

Complete pine scale



Brand roles



UI brand primary
#165E4A
Primary action in light surfaces.

Universal mark



Logo / mark
#25856A
Balanced brand mark across light and dark contexts. Not a default small-text color.

Secondary accent



Secondary brand
#2E8B6E
Supporting brand and decorative accent.

Warm accent



Warm emphasis
#D9A441
Sparse emphasis, highlights and selected data series.

Recommended visual balance



- Large surfaces stay neutral.
- Brand color establishes identity and primary actions.
- Semantic and data colors are used locally.

Core rules

- Do not use green as the only indication of success.
- Keep highly saturated colors for smaller elements.
- Use semantic tokens in components instead of raw HEX values.
- Do not reuse status colors as generic decoration.
- Color meaning must survive the monochrome theme.

Technical basis

Color space: sRGB
Specification: HEX / RGB
Accessibility target: WCAG 2.2 AA
Channels: Web, Desktop, CLI, PDF
Implementation: semantic design tokens

Token layers

Brand primitives

 color.brand.uiPrimary	#165E4A
 color.brand.mark	#25856A
 color.brand.secondary	#2E8B6E
 color.brand.warm	#D9A441

Interactive semantics

 color.action.primary.default	#165E4A
 color.action.primary.hover	#124E3D
 color.action.primary.active	#0D3E31
 color.focus.ring	#2F8F83

Theme semantics

 color.background.canvas	#F6FAF8
 color.background.surface	#FFFFFF
 color.text.primary	#18201D
 color.border.default	#C7D1CC

Reserved brand asset slots



Placeholders are intentional: approved artwork should be inserted without changing page structure or token definitions.

2A. Light theme

Clean surfaces, a subtle green canvas and high-contrast typography.

Illustrative UI example - Light theme (layout is not normative)

JELICA

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Projects

Analyze

Settings

Projects

New project

Search projects...

Name	Status	Updated
Project 1	Completed	12 May
Project 2	Running	11 May
Project 3	Failed	10 May

Theme tokens

 background-canvas	#F6FAF8	 text-disabled	#929D98
 background-surface	#FFFFFF	 text-inverse	#F7FBF9
 background-subtle	#EEF4F1	 text-brand	#165E4A
 background-muted	#E5ECE8	 text-link	#12634E
 background-elevated	#FFFFFF	 text-link-hover	#0A493A
 background-inverse	#18201D	 border-subtle	#E0E7E3
 background-code	#F0F4F2	 border-default	#C7D1CC
 text-primary	#18201D	 border-strong	#8C9993
 text-secondary	#46514C	 border-brand	#25856A
 text-muted	#66736D	 divider	#D9E1DD

Backgrounds

 background-canvas	#F6FAF8
 background-surface	#FFFFFF
 background-subtle	#EEF4F1
 background-muted	#E5ECE8
 background-elevated	#FFFFFF
 background-inverse	#18201D
 background-code	#F0F4F2

Text

 text-primary	#18201D
 text-secondary	#46514C
 text-muted	#66736D
 text-disabled	#929D98
 text-inverse	#F7FBF9
 text-brand	#165E4A
 text-link	#12634E
 text-link-hover	#0A493A

Borders

 border-subtle	#E0E7E3
 border-default	#C7D1CC
 border-strong	#8C9993
 border-brand	#25856A
 divider	#D9E1DD

Surface hierarchy

Canvas
#F6FAF8

Surface
#FFFFFF

Subtle
#EEF4F1

Inverse
#18201D

Usage note: The universal brand mark color #25856A is reserved for the logo/mark and selected borders. For small text, use theme-specific text tokens such as text-brand or text-link.

JELICA Color Design System

Light theme · 3/10

2B. Dark and monochrome themes

Dark uses graphite-pine surfaces; monochrome relies on structure, icons and patterns.

Illustrative UI example - Dark theme (layout is not normative)

JELICA

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Projects

New project

Search projects...

Name	Status	Updated
Project 1	Completed	12 May
Project 2	Running	11 May
Project 3	Failed	10 May

background-canvas	#0F1714	text-disabled	#65716B
background-surface	#16201C	text-inverse	#17201C
background-subtle	#1B2924	text-brand	#75BCA7
background-muted	#24332D	text-link	#7BCDB5
background-elevated	#202D28	text-link-hover	#A6E1D0
background-inverse	#EDF5F1	border-subtle	#283630
background-code	#0B1210	border-default	#394942
text-primary	#EDF5F1	border-strong	#62716A
text-secondary	#BDC9C3	border-brand	#43A086
text-muted	#8F9E97	divider	#303F39

Illustrative UI example - Monochrome theme (layout is not normative)

JELICA

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Settings

Projects

New project

Search projects...

Name	Status	Updated
Project 1	Completed	12 May
Project 2	Running	11 May
Project 3	Failed	10 May

background-canvas	#FFFFFF	text-muted	#666666
background-surface	#FFFFFF	text-disabled	#999999
background-subtle	#F4F4F4	border-default	#BDBDBD
background-muted	#E8E8E8	border-strong	#5E5E5E
text-primary	#111111	accent-primary	#111111
text-secondary	#444444	focus-ring	#000000

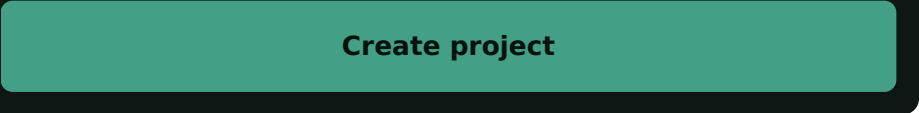
Not grayscale-only: status is reinforced with labels, icons, border styles, line weight and patterns.

Semantic equivalence between themes

Light - Primary action



Dark - Primary action



Monochrome - Primary action



Dark theme: avoid pure black for large surfaces. Use layered graphite-pine surfaces so borders, elevation and status callouts remain visible.

Monochrome theme: do not infer state from shade alone. Every meaningful state must retain a non-color cue.

3. Semantic colors

Status = color + icon + text. Color is never the sole carrier of meaning.

Light theme

Status	Accent	Surface	Text	Border
 Success	 #027A48	 #ECFDF3	 #05603A	 #6CE9A6
 Info	 #175CD3	 #EFF8FF	 #1849A9	 #84CAFF
 Warning	 #B54708	 #FFFAEB	 #93370D	 #FEC84B
 Error	 #B42318	 #FEF3F2	 #912018	 #FDA29B
 Progress	 #2563D9	 #EEF4FF	 #1849A9	 #8EB5FF
 Paused	 #6941C6	 #F4F3FF	 #53389E	 #BDB4FE
 Cancelled	 #59636E	 #F2F4F5	 #414A53	 #B8C0C7
 Neutral	 #66736D	 #F2F5F3	 #46514C	 #C7D1CC

Example: a validation error uses an error label, icon and border in addition to the red accent.

Dark theme

Status	Accent	Surface	Text	Border
 Success	 #32D583	 #102A20	 #75E0A7	 #087443
 Info	 #53B1FD	 #102A3D	 #84CAFF	 #1570BE
 Warning	 #FDB022	 #30240E	 #FEC84B	 #B54708
 Error	 #F97066	 #351512	 #FDA29B	 #B42318
 Progress	 #619BFF	 #132544	 #8EB5FF	 #2E62C4
 Paused	 #9B8AFB	 #241D3D	 #BDB4FE	 #6941C6
 Cancelled	 #98A2AB	 #252B2E	 #C4CBD1	 #59636E
 Neutral	 #8F9E97	 #20A26	 #BDC9C3	 #52615A

Dark surfaces: bright accents stay local; large status surfaces use muted background colors.

JELICA task state mapping

queued → neutral
running → progress

pause_requested → progress + waiting
paused → paused

cancel_requested → warning / neutral-progress

completed → success
failed → error
cancelled → cancelled

Status surfaces

Backgrounds are intentionally low saturation so text and icons remain dominant.

Do not overload green

Brand green and success green are separate roles. A green element is not automatically a success state.

Monochrome fallback

Success, warning and error remain distinguishable through symbol, label and border treatment.

4. Interactive states

Default, Hover, Active, Focus, Selected, Disabled.

Light theme

Primary button

Default	Hover	Active	Focus	Selected	Disabled
Button	Button	Button	Button	Button	Button
#165E4A #FFFFFF	#124E3D #FFFFFF	#0D3E31 #FFFFFF	#165E4A #FFFFFF	#103E32 #FFFFFF	#D7E1DC #87938D

Secondary button

Default	Hover	Active	Focus	Selected	Disabled
Button	Button	Button	Button	Button	Button
#FFFFFF #165E4A	#EFF8F4 #124E3D	#D7ECE5 #103E32	#FFFFFF #165E4A	#D7ECE5 #103E32	#F4F6F5 #929D98

Neutral row / list item

Default	Hover	Active	Focus	Selected	Disabled
Button	Button	Button	Button	Button	Button
#FFFFFF #18201D	#EEF4F1 #18201D	#DDEAE4 #18201D	#FFFFFF #18201D	#D7ECE5 #18201D	#F4F6F5 #929D98

Dark theme

Primary button

Default	Hover	Active	Focus	Selected	Disabled
Button	Button	Button	Button	Button	Button
#43A086 #07110D	#59B299 #07110D	#75BCA7 #07110D	#43A086 #07110D	#75BCA7 #07110D	#293A33 #69766F

Secondary button

Default	Hover	Active	Focus	Selected	Disabled
Button	Button	Button	Button	Button	Button
#16201C #75BCA7	#182924 #78CDB5	#24332D #A6E1D0	#16201C #75BCA7	#24332D #A6E1D0	#202D28 #8F9E97

Neutral row / list item

Default	Hover	Active	Focus	Selected	Disabled
Button	Button	Button	Button	Button	Button
#16201C #EDF5F1	#182924 #EDF5F1	#24332D #EDF5F1	#16201C #EDF5F1	#1C4437 #EDF5F1	#202D28 #657168

Focus: use a separate visible focus ring; do not make focus indistinguishable from hover or selected. Baseline recommendation: 2 px ring + 2 px offset.

Disabled: do not implement disabled state only with global opacity. Use explicit background, text and border tokens so readability stays predictable.

Keyboard first

All keyboard-operable controls need a visible focus state.

Selected ≠ Focus

Selection is persistent application state; focus is transient navigation state.

Pointer states

Hover and active states may reinforce interaction but cannot replace keyboard focus.

5. Data visualization palette

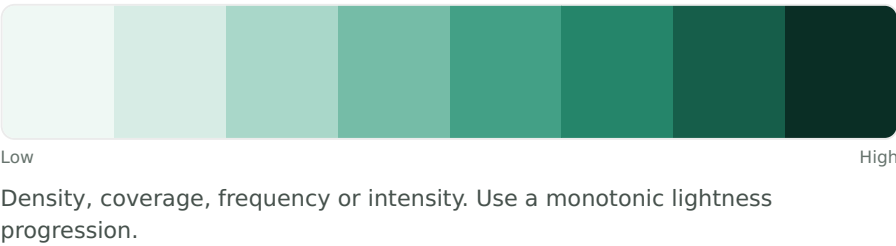
Categorical, sequential and diverging scales with non-color fallbacks.

Categorical palette - 10 colors

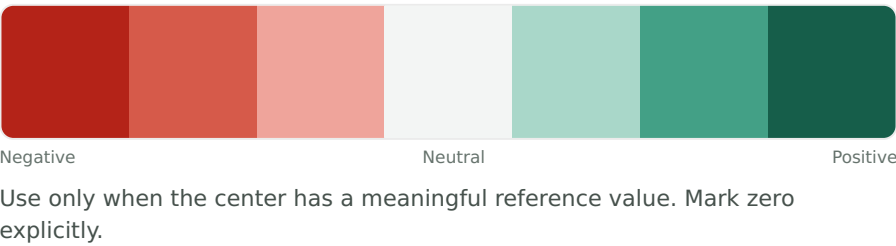


Recommended 4-5 series subset: #1B6F5A · #3A78B8 · #C19428 · #6A5FA8 · #C56A3A. Token numbers are stable identifiers; descriptive names are documentation labels only.

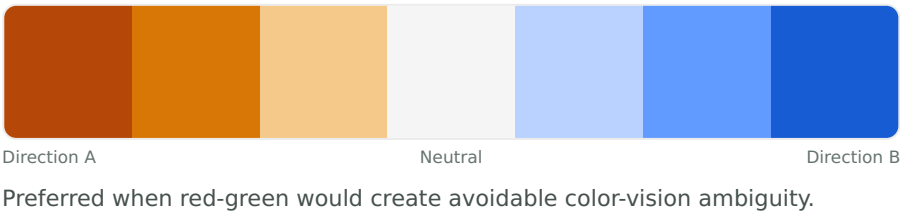
Sequential scale



Diverging scale



Alternative blue-orange



Encode important distinctions twice



Do not assign categorical colors semantic meaning. A series color should remain a series identity, not silently become “success”, “error” or “warning”.

Legends are mandatory whenever colors identify samples, clades, groups or categories. Dense charts should also use labels, marker shapes or patterns.

6. IUPAC nucleotide codes

Four canonical colors + compositional patterns + mandatory code letters.

Canonical bases

A
Adenine
#2E8B57

C
Cytosine
#2F6FDB

G
Guanine
#D99000

T
Thymine
#D1495B

U
Uracil
#D1495B

T and U share the same base color, but the letter remains visible so DNA/RNA notation is never hidden.

Two-base ambiguity codes

R
A/G
compact #878B2B

Y
C/T
compact #765FB0

S
G/C
compact #248B91

W
A/T
compact #B96D55

K
G/T
compact #C36C2F

M
A/C
compact #318F86

Patterns show composition in analytical mode; the compact fallback color is used when cells are too small for patterns.

Three-base ambiguity codes

B
C/G/T
compact #5B7182

D
A/G/T
compact #8F7637

H
A/C/T
compact #75677F

V
A/C/G
compact #477E70

Unknown and gap

N
A/C/G/T
#737B78

gap
#B5BBB8 / empty cell

Display modes

Compact mode: solid fill + code letter; ambiguity colors are muted; tooltip explains composition; gap is an empty cell with a line.

Analytical mode: two- and three-base codes use composite patterns; N uses a neutral dotted pattern; the code letter is always rendered above the pattern.

Monochrome fallback

A
solid

C
diagonal

G
dots

T/U
horizontal

Ambiguous codes combine the relevant base patterns. N uses a neutral dotted pattern and gap remains visually empty. The visible code letter is the primary identifier.

7. CLI color and symbol conventions

Standard ANSI colors with text-first semantics and optional emoji.

```
🌲 Analysis task 1bd68f84... was created and started.
✓ Configuration parsed successfully.
i Input records validated.
! Ambiguous symbols detected in 2 records.
● Alignment stage: 4 / 7 records processed.
⏸ Task paused at a safe checkpoint.
- Cancelled task retained for inspection.
✖ Validation failed: unsupported input format.

No-color fallback:
[INFO] Input records validated.
[WARN] Ambiguous symbols detected.
[ERROR] Validation failed.
```

Rule: ANSI color supplements words, status labels and symbols. Output must remain understandable when color is disabled or unavailable.

ANSI mapping

Role	SGR	ANSI color	Usage
 Brand / normal success	32	Green	Used with explicit text/symbols
 Information	36	Cyan	Neutral informational output
 Warning	33	Yellow	Potential issue or user attention
 Error / failed	31	Red	Failure or invalid input
 Progress / running	34	Blue	Active task or stage progress
 Paused	35	Magenta	Paused state
 Neutral / cancelled	90	Bright black / gray	Muted or terminal-neutral state
 Default text	39	Terminal foreground	Normal explanatory text

Use the terminal default foreground/background whenever possible. Do not require 256-color or true-color support for meaning.

Default identity marker

 Evergreen Tree emoji accompanies selected high-level CLI messages and reinforces the JELICA identity.

Seasonal Easter egg

 Christmas Tree replaces the default tree on Dec 24-26 and Jan 6-8. It is decorative and never carries state meaning.

User control

Color and emoji are optional. CLI settings may disable them independently. Plain-text status labels remain the accessibility and compatibility fallback.

Cross-platform constraint: ANSI semantics must be consistent on macOS, Linux and modern Windows terminals; exact rendered hues may vary by terminal theme.

Do not mirror Web HEX values in ANSI. Terminal colors are semantic aliases, not exact brand-color reproduction.

8. Accessibility and implementation principles

WCAG 2.2 AA rules that directly shape the JELICA color system.

SC 1.4.1 · Level A

Use of Color

Color must not be the only visual way to convey information, indicate an action, request a response or distinguish an element. JELICA therefore pairs status color with text, icons, patterns, markers or border treatment.

SC 1.4.3 · Level AA

Contrast (Minimum)

Normal text requires at least **4.5:1** contrast against its background. Large text requires at least **3:1**. Logotypes are exempt, but interactive uses of a logo should still remain clearly visible.

SC 1.4.11 · Level AA

Non-text Contrast

Visual information needed to identify UI components, their states, and essential graphical objects should reach at least **3:1** contrast against adjacent colors.

SC 2.4.7 / 2.4.11 · Level AA

Keyboard Focus

Keyboard focus must be visible, and author-created content must not completely hide the focused component. JELICA uses a distinct focus ring rather than relying on hover or selected styling.

Verified contrast examples from this palette

Light body text

15.79:1

#18201D on #F6FAF8

Dark body text

16.42:1

#EDF5F1 on #0F1714

Light primary button

7.68:1

#FFFFFF on #165E4A

Dark primary button

6.04:1

#07110D on #43A086

Important brand-mark distinction: #25856A is a universal logo/mark color, not a general-purpose small-text token. Its contrast is about **4.30:1** on the light canvas and **4.03:1** on the dark canvas, so theme-specific text tokens must be used for small text.

Internal design rule: treat WCAG thresholds as minimums, not visual targets. Prefer clear margin above the threshold, especially for small text, thin strokes, dense tables and status-heavy analytical views.

Color-vision robustness

Never require users to distinguish red from green, or any other hue pair, without a second cue. Test charts and status components under common color-vision simulations.

Monochrome parity

Meaning must survive grayscale/monochrome output through text, symbol, line style, marker shape and pattern.

Implementation discipline

Components consume semantic tokens. Theme files map semantics to actual colors. Raw HEX values should stay in the token layer, not in component logic.

W3C references

WCAG 2.2 Understanding Docs: SC 1.4.1 Use of Color · SC 1.4.3 Contrast (Minimum) · SC 1.4.11 Non-text Contrast · SC 2.4.7 Focus Visible · SC 2.4.11 Focus Not Obscured (Minimum). This page is a design-system-oriented summary, not a complete WCAG conformance checklist.